

Requirement evidence snapshot (see CSV for full matrix)

id	status	confidence	predicted
R01_liquid_delivery	PASS	high (by design; analogue Chen 2023)	liquid emulsion, no solids injected
R02_viscosity	PASS	medium-high at 20-60 C; DCMO 1 cP is published	0.64 (nominal 60 C)
R03_transport_temperature	UNKNOWN	low at 150 C	models evaluated to 150 C
R04_transport_pressure	UNKNOWN	low (pressure effect on RORP not retrieved as a quantitative rate law)	10000 psi = 68.90 MPa; liquids remain liquid
R05_post_transform_pressure	PASS	medium at 60 C; low at 150 C	6000 psi = 41.369 MPa
R06_controlled_activation	PASS	medium (published latency analogues)	temperature + inhibitor index
R07_transformation_window	MARGINAL	medium in 40-80 C; low/UNKNOWN at 150 C	78.4 min nominal; HWPT 1.6 min
R08_discrete_particles	PASS	medium (analogue beads; not PhaseForge experiments)	pH=0.85 continuous; discrete if pH<0.55
R09_particle_size	PASS	medium (order of magnitude; correlation uncertainty)	273
R10_sphericity	MARGINAL	medium analogue; not measured here	near-spherical expected if IPT stabilised
R11_minimal_adhesion	PASS	low	0.27
R12_load_bearing	PASS	medium at T<=Tg; UNKNOWN near Tg	SF=1.41 at nominal T
R13_compressive_integrity	PASS	low at 150 C	SF=1.41 (nom); HWPT SF=0.33
R14_remain_distributed	MARGINAL	low-medium	liquids can enter branches (Chen analogue)
R15_open_pathways	PASS	medium if discrete; FAIL if bulk gel	porosity=0.85
R16_flow_through_around	PASS	medium	0.759
R17_no_bulk_gel	MARGINAL	low-medium	PASS only if droplets remain isolated until solid